Local Deployment and Configuration of Elastic SIEM with Sysmon Integration

Introduction

This report presents the deployment and configuration of a local Elastic SIEM environment. The project involves installing the Elastic Stack (Elasticsearch and Kibana), configuring Sysmon to monitor system-level events, and integrating Elastic Agent to collect and forward logs. The purpose of this setup is to enable real-time visibility into system activities and support basic security monitoring and analysis.

Executive Summary

A local Security Information and Event Management (SIEM) system was successfully deployed using the Elastic Stack. The environment enables centralized log collection and monitoring of system activities.

Sysmon was used to generate detailed system logs, including process creation and network activity, while Elastic Agent was configured to forward these logs to the SIEM platform. The collected data was visualized and analyzed using Kibana.

The implemented solution demonstrates the ability to monitor system behavior and detect potential suspicious activities. This improves the overall security visibility of the system.

Future improvements may include enabling alerting rules, integrating additional log sources, and enhancing log analysis capabilities.

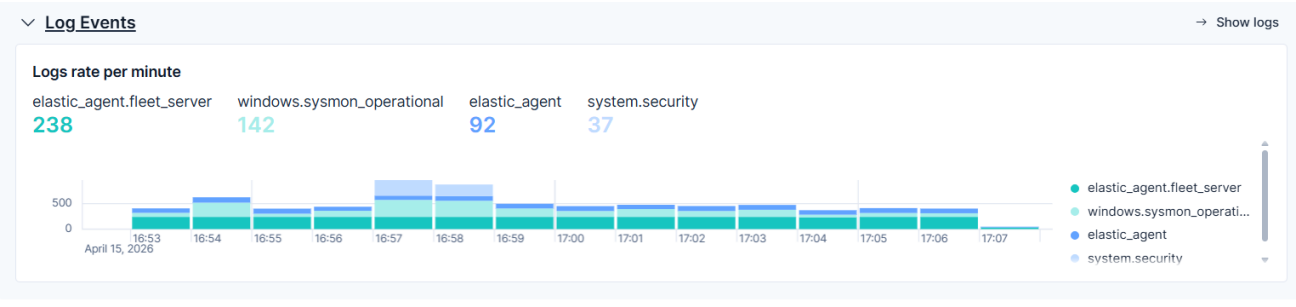


Figure 1: Real-time log ingestion distribution by source (Sysmon, Security, and Agent health)

Technical Report

1. Target & Environment Overview

Feature	Details
Hostname	MSI
IP Address	26.71.172.34
SIEM Platform	Elastic Stack 9.3.3(localhost:5601)
Components	Elasticsearch, Fleet-Managed Elastic Agent
Logging Tool	Microsoft Sysmon v15.20
Status	Monitored

2. Implementation Steps

2.1 Elastic Stack Installation

The Elastic Stack (Elasticsearch and Kibana) was installed locally on a Windows machine to create the SIEM environment.

1. Download Elasticsearch and Kibana from the official Elastic website.

    - Elasticsearch (Windows): https://www.elastic.co/downloads/elasticsearch
    - Kibana (Windows): https://www.elastic.co/downloads/kibana

2. Extract both packages into local directories.
3. Open Command Prompt and navigate to the Elasticsearch directory.
4. Run the following command to start Elasticsearch: `elastic.bat`
5. Wait until the service starts and note the generated credentials (username and password).
6. Open another Command Prompt window and navigate to the Kibana directory.
7. Start Kibana using: `kibana.bat`
8. Open a web browser and access Kibana:

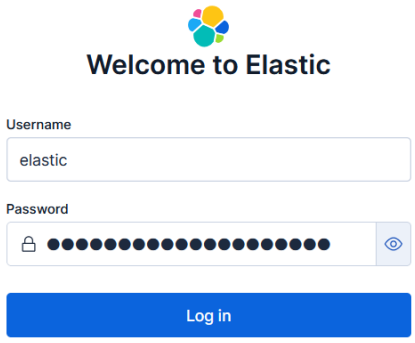


Figure 2: Kibana login interface for secure administrative access to the SIEM dashboard.

9. Complete the setup using the enrollment token and login credentials.

2.2 Fleet Configuration

To enable centralized management of Elastic Agents, the **Fleet** server was initialized using the following steps:

1. Navigate to **Management > Fleet** in the Kibana console.
2. Select **Add Fleet Server** and follow the on-screen instructions to generate the deployment command.

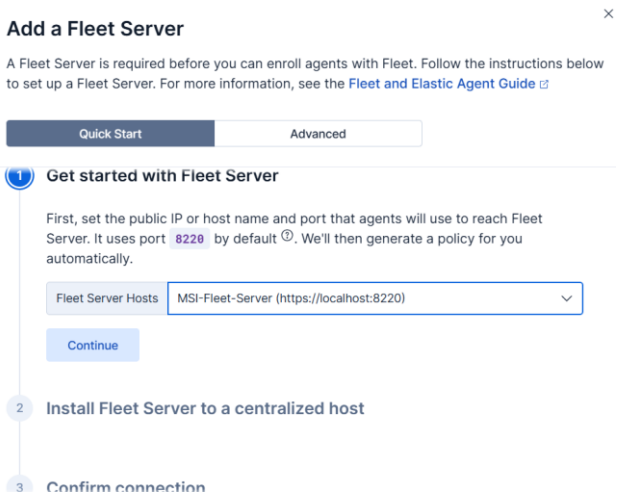


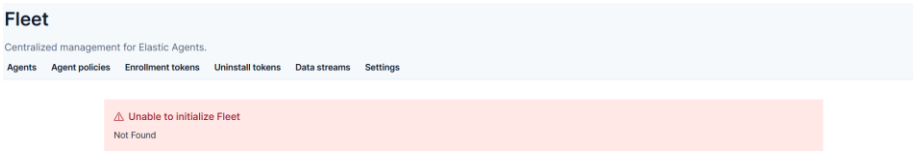
Figure 3: Initializing the Fleet Server setup

3. Run the generated command on the target host to install and connect the Fleet Server.
4. Verify the Fleet Server status is "Healthy" in the Kibana dashboard.

Note: Troubleshooting Fleet

If initialization fails, add a 32-character key to **kibana.yml**:
`xpack.encryptedSavedObjects.encryptionKey: "32_char_key_here"`

Then, **restart Kibana** to apply the fix and stabilize the service.



2.3 Sysmon Installation

Sysmon was installed to generate detailed Windows event logs.

1. Download & Extract: Get Sysmon from Microsoft Sysinternals and extract the archive.
2. Installation: Run `Sysmon64.exe -i` via Administrator Command Prompt.
3. Verification: Confirm the service is active using `sc query sysmon64`.

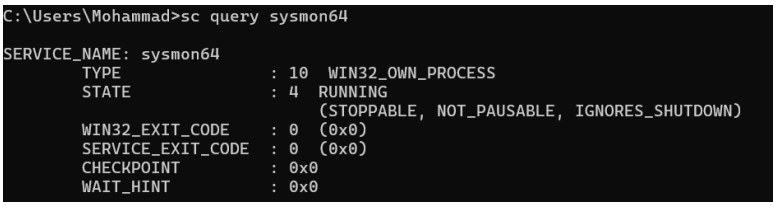


Figure 4: Confirming successful Sysmon64 installation and service activation.

4. Log Observation: View generated events in Event Viewer under Sysmon > Operational.

2.4 Elastic Agent Installation

Elastic Agent was installed and enrolled to forward logs to the SIEM platform.

1. Enrollment: In Kibana, go to **Fleet > Add Agent** and copy the install command.

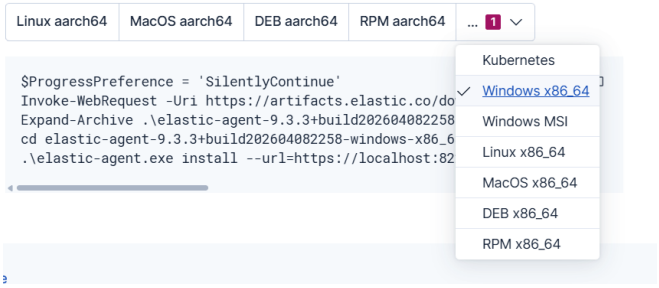


Figure 5: Selecting the Windows x86_64 Elastic Agent binary and installation command.

2. Deployment: Execute the command in **PowerShell (Admin)** to install and enroll the agent.
3. Validation: Check the **Fleet** dashboard to verify the agent's presence.



Figure 6: Successful agent enrollment verification

Log Collection and Analysis

After completing the setup, logs were successfully collected and displayed in Kibana. The system captured multiple types of events, including:

- Process creation events
- Network activity
- General system logs

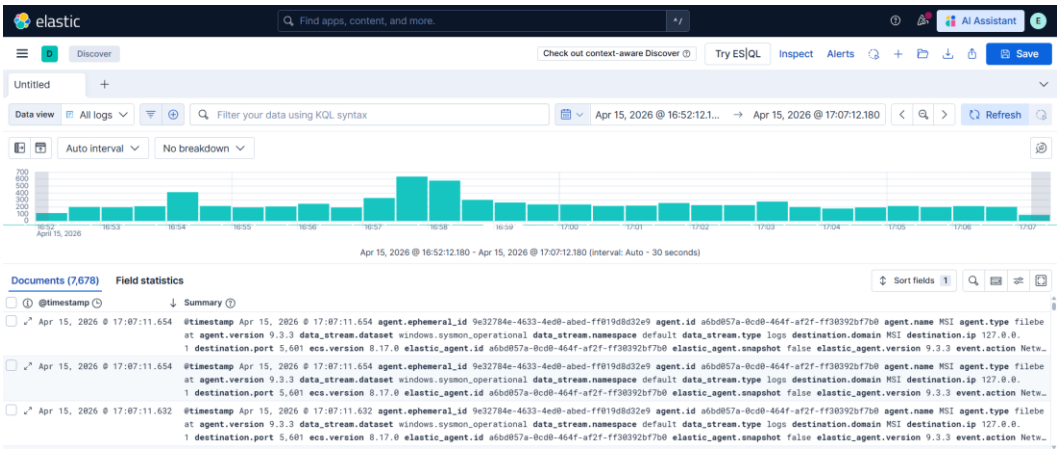


Figure 7: Kibana Discover tab displaying indexed logs with detailed metadata and event timestamps.

Proof of Concept

To verify the system functionality, test activities were performed such as launching applications and generating network traffic. These actions were successfully captured by Sysmon and appeared in Kibana logs, confirming proper integration between all components.

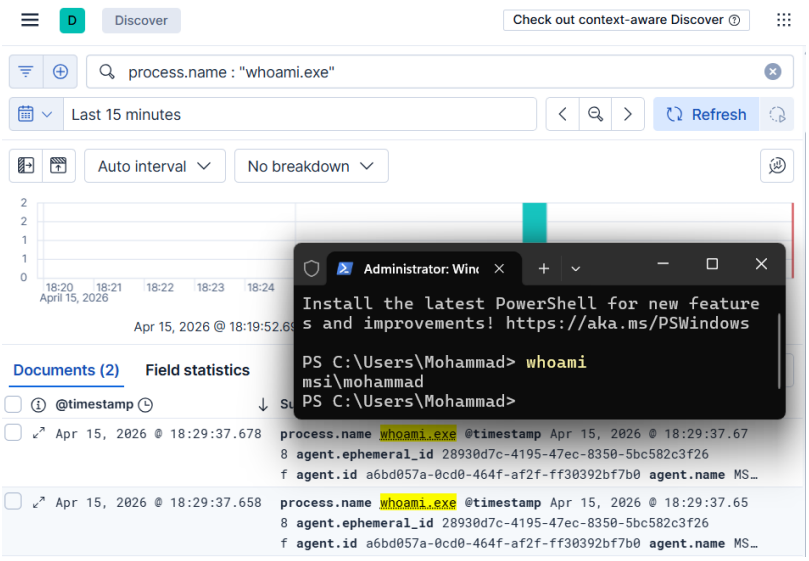


Figure 8: End-to-end validation showing command execution in PowerShell and its corresponding indexed log in the SIEM.

Recommendations

- Enable alerting features in Kibana for real-time threat detection
- Integrate additional log sources for better visibility
- Regularly update Sysmon configuration for improved logging
- Monitor logs continuously to detect anomalies

Conclusion

The deployment successfully established a centralized SIEM environment using the Elastic Stack and Fleet-managed Agents. By integrating Microsoft Sysmon, the system achieved deep endpoint visibility, enabling real-time detection of system-level activities. The Proof of Concept (PoC) demonstrated the platform's capability to ingest, parse, and visualize security events (such as user discovery commands) within seconds. This infrastructure provides a scalable foundation for advanced threat hunting and continuous security monitoring.